Prabhavit Gupta

AI Engineer | Threat Intelligence Agents, Multi-Agent Systems & Cloud GenAI Infrastructure

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Executive Summary

- **Applied AI & Multi-Agent Engineering:** Software engineer with 3 years commercial experience architecting and shipping autonomous agent systems, multi-agent crews (LangGraph, CrewAI-style loops), and production LLM pipelines. Hands-on experience developing agentic workflows to analyze high-volume semi-structured telemetry, summarize anomalous signals, and accelerate technical investigations.
- **Model Context Protocol (MCP) & API Security:** Specialist in Model Context Protocol (MCP Server & Client SDKs) and agent-to-API communication security. Experienced implementing credential sandboxing, payload sanitization, least-privilege tool execution, and guardrails to mitigate prompt injection, malicious agent tool calls, and data exfiltration.
- **Cloud Infrastructure & Eval-Driven Quality:** Proficient building and deploying containerized Python microservices (FastAPI, Docker) across AWS and GCP. Experienced establishing automated regression test harnesses (RAGAS, synthetic incident datasets) and distributed tracing (Langfuse, OpenTelemetry) to benchmark detection precision, latency, and token economics.

Technical Skills & Core Competencies

- **Applied AI & Multi-Agent Systems:** Multi-Agent Crews, Autonomous Tool Loops, CrewAI / LangGraph Orchestration, Threat Analysis Automation, State Machine Persistence, Dynamic Context Pruning.
- **Protocols & API Security:** Model Context Protocol (MCP), REST / WebSocket APIs, API Security, Prompt Injection Mitigation, Sandboxed Tool Execution, Guardrails (NeMo / Llama-Guard), Audit Logging.
- **Cloud & Data Infrastructure:** Python (AsyncIO, FastAPI, PyTest, Pydantic), AWS (Bedrock, S3, ECS), GCP (Vertex AI), PostgreSQL (pgvector), Redis (Caching & Semantic Queues), Docker.
- **Evaluation & Observability:** Eval-Driven Development, RAGAS Benchmarking, Synthetic Test Dataset Generation, Langfuse Distributed Tracing, Latency Profiling, Token Budgeting, CI/CD.
- **Engineering Practices:** Git, GitHub Actions, Defensive Coding, Architecture Decision Records (ADRs), Rapid Prototyping & Hacker Mentality.

Professional Experience

AI Engineer & Solutions Consultant • *Independent / AI Product Engineering Consultancy*
2024 – Present • Venice, IT (Remote Europe)

- Architected and deployed production agentic systems and tool-using LLM platforms for European tech clients, translating complex domain specifications into high-performance cloud architectures.
- Developed collaborative multi-agent crews coordinating specialized analysis nodes to ingest, cross-reference, and summarize high-volume event signals and suspicious activity logs.
- Standardized secure tool interfaces using Model Context Protocol (MCP), enforcing rigorous schema validation and permission isolation between autonomous agents and backend databases.
- Designed automated evaluation test suites with RAGAS and synthetic adversarial queries, continuously benchmarking agent tool-calling accuracy, hallucination resistance, and latency.
- Integrated real-time distributed tracing using Langfuse and OpenTelemetry, establishing operational visibility into decision trees, error recovery paths, and token consumption.
- Packaged AI services into lightweight Docker containers and automated regression validation within GitHub Actions CI/CD workflows, ensuring stable, reproducible cloud deployments.

Full-Stack AI Software Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Engineered high-throughput asynchronous backend pipelines in Python and TypeScript for parsing, indexing, and classifying real-time streaming data.
- Implemented semantic search and embedding retrieval using PostgreSQL (pgvector) and Redis, delivering reliable sub-second response times for downstream automated classifiers.
- Developed comprehensive test automation with PyTest and integrated automated quality gates into CI/CD pipelines to guarantee production stability.
- Collaborated with cross-functional technical teams to convert threat intelligence and data governance requirements into resilient, modular microservices.

Key Engineering Projects

- **Autonomous Threat Analysis Multi-Agent Crew (Python + LangGraph + MCP):** Multi-agent collaborative system coordinating automated incident log parsing, threat artifact extraction, and summarized reporting with Langfuse distributed tracing.
- **High-Throughput Semantic Verification Engine (FastAPI + pgvector + Redis):** Sub-second embedding retrieval and payload classification microservice with Pydantic structured output validation, token budgeting, and automated evaluation harnesses.

Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy & Economics** – Ca' Foscari University of Venice (2018 – 2022)
  Rigorous foundation in formal logic, statistical reasoning, decision theory, and complex systems analysis.
- **Languages:** English (C1 Full Professional / Fluent), Italian (Working proficiency), French (Basic understanding).
- **Work Authorization & Legal Status:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy) – Unrestricted legal right to live and work across the EU without sponsorship; available immediately for remote engagement.